



Service-Checklist

Name:	Date:	
		Work order no.
Service Job at:		
Company		
Address		
	Street / City / Country	

!! For all systems to be filled in !!

1. System and information about situation

1.1 System description:

HIMA-Job-Nr.:	i.e. 6035043 Pos. 1		
Process name:	site name, used at customers, i.e. HDPE		
System:	☐ HIQUAD	☐ HIMatrix	☐ HIMax ☐ others: _____
Programming tool:	i.e. ELOP II V3.5 (Build 6812) – for ELOP II Factory please support Build of Project- AND Hardwaremanagement (via Info) !		
Operation system CPU:	i.e. BS51q V7.0-7 (0337)	Operation system COM:	ONLY for HIMatrix i.e. COM V4.24
Communications-Mod:	i.e. F8621A	No. / pos.	i.e. V6.0-6 (9808)
Communications-Mod:	i.e. F8621A	No. / pos.	i.e. V6.0-6 (9808)
Communications-Mod:	i.e. F8621A	No. / pos.	i.e. V6.0-6 (9808)
Compiler (only ELOP):	i.e. C50-H V4.1-2 (9028)		
Wired by	Company / City / contact (if known)		

1.2 Fault description on arrival:

i.e. Read-out of diagnostic display at H41q/H51q

1.3 Measures / investigations, already done by customer:

i.e. New Start done

!! For all systems to be filled in !!

2. General test protocol

All procedures will have the aim to reproduce the reason for the disturbance, if possible.

Before start of investigations (before any manipulations are done) at first contents of all diagnostics (i.e. detailed error list) has to be noticed, copied as a file or printed out (for redundant systems it has to be done for both CPUs).

2.1 Testing of 24V supply voltage

- | | |
|---|--|
| <ul style="list-style-type: none"> ● Is 24V supply voltage redundant? ● Measurement of voltage on input terminals ! | <div> ☐Yes ☐No </div> <div> $\frac{U_1 = \quad}{\quad} V$ $\frac{U_2 = \quad}{\quad} V$ </div> |
| <ul style="list-style-type: none"> ● Are all connections tight in the terminals (pull the wire, before tightening the screws again)? | <div> ☐Yes ☐No </div> <div> remark _____ </div> |
| <ul style="list-style-type: none"> ● Is no isolation inside terminal contacts? | <div> ☐Yes ☐No </div> <div> remark _____ </div> |
| <ul style="list-style-type: none"> ● Is crimping done correctly? | <div> ☐Yes ☐No </div> <div> remark _____ </div> |
| <ul style="list-style-type: none"> ● For parallel 24V supply: Is decoupling ok? | <div> ☐Yes ☐No </div> <div> remark _____ </div> |
| <ul style="list-style-type: none"> ● Where is 24V supply coming from? | <div> remark _____ </div> |
| <ul style="list-style-type: none"> ● Is an electric rectifier existing? | <div> ☐Yes ☐No </div> <div> remark _____ </div> |
| <ul style="list-style-type: none"> ● Is the supply by a HIMA 24V power module? | <div> ☐Yes ☐No </div> <div> Power module type _____ </div> |
- i.e. PS1000

2.2 Testing of 24V decoupling and power distribution

- | | | |
|---|------------------------------|-----------------------------|
| ● Which assembly kit is used? | i.e. K7213 | |
| ● Measuring of EL+ voltage! | U = _____ | V |
| ● Measuring of L+ voltage! | U = _____ | V |
| ● Are all EL+, L+ and L- contacts tight at the decoupling module (pull the wire, before tightening the screws again)? | ☐ Yes | ☐ No |
| | remark | _____ |
| ● Is no isolation inside contacts? | ☐ Yes | ☐ No |
| | remark | _____ |
| ● Is crimping done correctly? | ☐ Yes | ☐ No |
| | remark | _____ |

!! For H41/H51 or H41q/H51q to be filled in !!

2.3 Measuring of 24V voltage at the central rack (not for HIMatrix)

- | | |
|--|--|
| <ul style="list-style-type: none"> Measuring of 24V voltage at power module (i.e. F7126) Nr1! | <div>U = <input type="text"/></div> <div>V</div> |
| <ul style="list-style-type: none"> Measuring of 24V voltage at power module (i.e. F7126) Nr2! | <div>U = <input type="text"/></div> <div>V</div> |
| <ul style="list-style-type: none"> Measuring of 24V voltage at power module (i.e. F7126) Nr3! | <div>U = <input type="text"/></div> <div>V</div> |
| <ul style="list-style-type: none"> Are all Faston crimps tight at the contact points (pull the wire, before tightening the screws again)? | <div> ☐Yes ☐No </div> <div> remark <input type="text"/> </div> |
| <ul style="list-style-type: none"> Is crimping of the Faston crimps done correctly? | <div> ☐Yes ☐No </div> <div> remark <input type="text"/> </div> |

2.4 Measuring of 24V voltage at the I/O racks (not for HiMatrix)

- | | | |
|--|------------------------------|-----------------------------|
| ● Measuring of 24V voltage at all contacts of (i.e.) F7133! | $U_{\max} =$ | V |
| | $U_{\min} =$ | V |
| ● What power distribution modules are in use ? | <hr/> i.e. F7133 | |
| ● Are all Faston crimps tight at the EL+ / L-contact points of F7133 (pull the wire, before tightening the screws again)? | ☐ Yes | ☐ No |
| | remark | <hr/> |
| ● Is crimping of the Faston crimps done correctly? | ☐ Yes | ☐ No |
| | remark | <hr/> |
| ● Measuring of 24V voltage on the back plane (if necessary, analogue modules)? | $U =$ | V |
| ● Are all Faston crimps tight at the contact points of the back plane (pull the wire, before tightening the screws again)? | ☐ Yes | ☐ No |
| | remark | <hr/> |
| ● Is crimping of the Faston crimps done correctly? | ☐ Yes | ☐ No |
| | remark | <hr/> |

2.5 Testing of 24V voltage at Group Shut Down (mostly only for old systems)

- | | |
|--|---|
| <ul style="list-style-type: none"> When using Group Shut Down modules is the wiring and voltage at the group relays ok? | ☐ Yes ☐ No
remark _____ |
| <ul style="list-style-type: none"> Is crimping of the cable endings at the group relays done correctly? | ☐ Yes ☐ No
remark _____ |
| <ul style="list-style-type: none"> Measuring of 24V voltage at all contact points of Group Shut Down! | $\underline{U_{\max} = \quad \quad \quad} \text{ V}$ $\underline{U_{\min} = \quad \quad \quad} \text{ V}$ |
| <ul style="list-style-type: none"> Which Group Shut Down module is used ? | |



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2.6 Testing of 5V voltage (not for HIMatrix)

- Measuring of 5V voltage on front of power module F7126! 5,4V requested, difference $\leq 0,025V$ between 2 power modules allowed

power module 1	power module 2	power module 3
$U_{ist} =$ _____ V	_____ V	_____ V
- Measuring of 5V voltage on front of power module F7126 at additional power supply rack!

power module 1	power module 2	power module 3
$U_{ist} =$ _____ V	_____ V	_____ V
- Measuring of 5V voltage (Gnd) on rear side of central rack!
(must be: $\geq 4,9 V$) $U_{ist} =$ _____ V
- Is the feeler line connected correctly? (for H50) ☐ Yes ☐ No

remark _____
- Are all Faston crimps (5 V area) tight at the contact points of the back plane (pull the wire, before tightening the screws again)? ☐ Yes ☐ No

remark _____
- Is crimping of the Faston crimps done correctly? ☐ Yes ☐ No

remark _____
- Measuring of 5V voltage on all I/O racks on slot 1 (right) (z6/d6 – z30/d30)!
(must be $\geq 4,9 V$)

cabinet 1		cabinet 2		cabinet 1		cabinet 2	
I/O rack	U_{alt}	I/O rack	U_{alt}	I/O rack	U_{alt}	I/O rack	U_{alt}
1	V	1	V	1	V	1	V
2	V	2	V	2	V	2	V
3	V	3	V	3	V	3	V
4	V	4	V	4	V	4	V
5	V	5	V	5	V	5	V
6	V	6	V	6	V	6	V
7	V	7	V	7	V	7	V
- Testing GND connection. Are all Faston crimps tight at the contact points of the back plane (pull the wire, before tightening the screws again)? ☐ Yes ☐ No

remark _____
- Is crimping of the Faston crimps done correctly? ☐ Yes ☐ No

remark _____
- Is 5V wiring be done in star form and redundant (acc. HIMA-Standard)? ☐ Yes ☐ No

remark _____
- Is Gnd wiring be done acc. HIMA-Standard? ☐ Yes ☐ No

remark _____



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2.7 General testing

● Are all IO-Bus connection cable be connected correctly and are the cable plug fixed in position?	☐ Yes	☐ No
	remark	_____
● Are the fans running?	☐ Yes	☐ No
	remark	_____
● Are air circulation ways disturbed or are distances too small?	☐ Yes	☐ No
	remark	_____
● Is there an earth fault detection to monitor short cuts?	☐ Yes	☐ No
	remark	_____
● Are all earthing contacts (cabinet, field cables, Bus, etc.) done correctly?	☐ Yes	☐ No
	remark	_____
● Is L- connected to earth?	☐ Yes	☐ No
	remark	_____
● Is there a connection between L- and Gnd?	☐ Yes	☐ No
	remark	_____
● Measuring of voltage between L- and 5V+!	U = _____	V
● Measuring of voltage between Gnd and 24V+!	U = _____	V
● Are all fixing screws of I/O modules tightened?	☐ Yes	☐ No
	remark	_____
● Are there more than 10 output modules within one I/O rack and has the coupling module F7541 release <05 or are there modules F7536/F7537 in use? (not for HIMatrix)	☐ Yes	☐ No
	remark	_____
● Are the bus terminators Z6007 or. F7546 correctly in position? (not for HIMatrix)	☐ Yes	☐ No
	remark	_____
● Are all communications line placed correctly and are all cable plugs fixed in position?	☐ Yes	☐ No
	remark	_____
● Are all I/O connection cables at HIMatrix tightened (pull the wire, before tightening the screws again)?	☐ Yes	☐ No
	remark	_____
● Is the crimping of the IO connection lines for HIMatrix done correctly?	☐ Yes	☐ No
	remark	_____
● Is the installation of HIMatrix modules ok?	☐ Yes	☐ No
	remark	_____
● Do all wirings have the correct diameter?	☐ Yes	☐ No
	remark	_____



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2.8 General testing

- Are the fans running?

☐ Yes

☐ No

remark _____

- Are air circulation ways disturbed or are distances too small?

☐ Yes

☐ No

remark _____

- Are all earthing contacts (cabinet, field cables, Bus, etc.) done correctly?

☐ Yes

☐ No

remark _____

- Is L- connected to earth?

☐ Yes

☐ No

remark _____

- Are all communications line placed correctly and are all cable plugs fixed in position?

☐ Yes

☐ No

remark _____

- Are all I/O connection cables at HIMatrix tightened (pull the wire, before tightening the screws again)?

☐ Yes

☐ No

remark _____

- Is the crimping of the IO connection lines for HIMatrix done correctly?

☐ Yes

☐ No

remark _____

- Is the installation of HIMatrix modules ok?

☐ Yes

☐ No

remark _____

- Do all wirings have the correct diameter?

☐ Yes

☐ No

remark _____



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3. Surrounding and external influences

- Common situation of system (i.e. very dirty etc.)? remark _____

- Is surrounding very warm or is there very high humidity? remark _____

- Exist other external influences (i.e. vibrations)? remark _____

- Additional remarks?

4. Other tests or measurements:

5. started measures to solve the problem / erase the reason:

i.e. change of a module, new wiring,

6. Final conclusion:

- Has the problem been solved? ☐ Yes ☐ No
- Was the customer been satisfied? ☐ Yes ☐ No
- Further measures / service jobs necessary? ☐ Yes ☐ No